

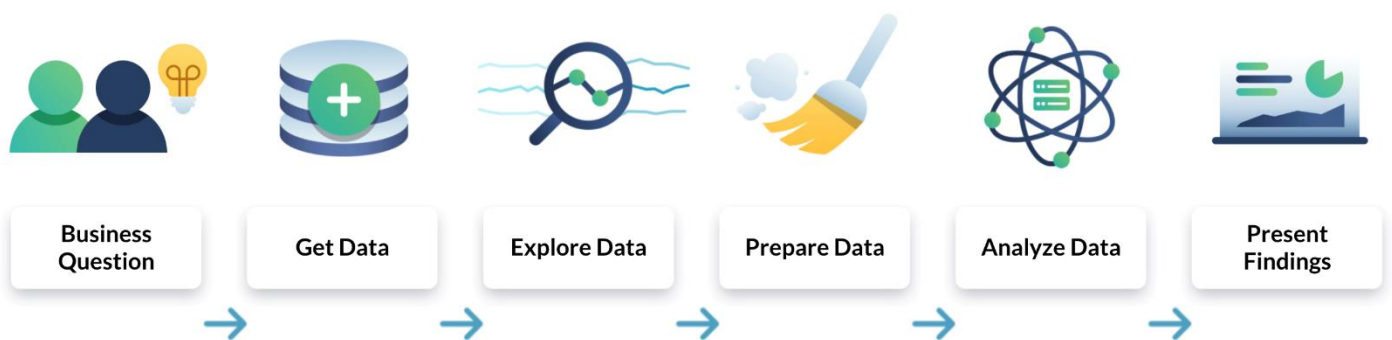
DATA ANALYST ROADMAP - 2024

Start your career to data science in just **3 months**, this roadmap will help you to learn data science skills from **scratch**. This is a structured and detailed roadmap which includes:

tech skills + resources, projects, soft skills + resources, resume template + tips and interview preparation guide

➤ Role of a Data Analyst

A data analyst collects, cleans, and interprets data sets in order to answer a question or solve a problem



Six steps of data analysis:

- 1. Business Question:** Define what problem you want to solve
- 2. Get Data:** Collect the data required for analysis
- 3. Explore Data:** Explore data with visual exploration to understand what is in a dataset
- 4. Prepare Data:** Data cleaning, calculated fields and data validation
- 5. Analyze Data:** Use data analysis techniques to understand, interpret, and derive conclusions based on the requirements
- 6. Present Findings:** Share insights with stakeholders

< In just 3 months you can **START** your career to data science >

Let's Start! ▶

1. Excel

- Topics
 - Basic formulas: SUM, AVERAGE, MEAN, MEDIAN, SUMPRODUCT, CONCATENATE
 - Advance formulas: VLOOKUP, INDEX, MATCH, IF, COUNTIF, SUMIF
 - Remove duplicates and conditional formatting
 - Charts, filters, sort and slicers
 - Pivot tables and pivot charts
 - Ignore VBA, Macros, etc
- Resources
 - Complete Excel Tutorial in one video: <https://www.youtube.com/watch?v=OX-iyb-21tk>
 - Excel for beginner's playlist:
<https://www.youtube.com/playlist?list=PLdOKnrf8EcP1Y1XRUVSUC0g-WfeiBigbd>
 - Google Template Gallery: <https://docs.google.com/spreadsheets/u/0/?ftv=1>
 - Full Excel Project: <https://www.youtube.com/watch?v=gTK5rNhWJyA>

2. Math and Statistics

- Topics
 - Basic Math: Arithmetic, Weighted average, Cumulative sum, Percentile
 - Basic Statistics: Mean, Median, Mode, Standard deviation, Normal distribution
- Resources
 - Note: just learn basics, initially don't go for phd 😊
 - Complete Statistics For Data Science In 6 hours By Krish Naik (watch only above mentioned topics): <https://www.youtube.com/watch?v=LZzq1zSL1bs>
 - Statistics Tutorial for Beginners by Simplilearn:
<https://www.simplilearn.com/tutorials/statistics-tutorial>

NEXT STEPS

- Create a professional LinkedIn profile
 - Add professional photo, headline, summary and educational details
- Make full Excel project and add to your resume/LinkedIn profile
- Optional: create a GitHub account

Hum honge kaamyaab.. 20% completed! 🚀

Week – 3, 4, 5

3. SQL

- Topics
 - Basic Queries: SELECT, WHERE, DISTINCT, LIKE, BETWEEN, ORDER BY, LIMIT, GROUP BY, HAVING CLAUSE, INSERT, UPDATE, ALTER, IMPORT, Data types
 - Advance Queries: Date time function, Window function, Sub query, Case statement, CTE, query optimisation
 - JOINS: Inner, Outer, Left, Right
- Resources to learn SQL
 - SQL tutorial in one video: <https://www.youtube.com/watch?v=On9eSN3F8w0>
 - SQL for beginner's playlist: https://www.youtube.com/playlist?list=PLdOKnrf8EcP17p05q13WXbHO5Z_JfXNpw
 - W3schools website to learn SQL: <https://www.w3schools.com/sql/>
 - SQL Data Analysis Portfolio Project: <https://www.youtube.com/watch?v=VFfUljswMKM>
 - SQL interview question's playlist: https://www.youtube.com/playlist?list=PLdOKnrf8EcP1y_LPEv7uBpzoRmlATjCVr

- Resources to practice SQL
 - W3schools: <https://www.w3schools.com/sql/>
 - Hacker rank sql: <https://www.hackerrank.com/domains/sql>
 - 8-week sql challenge- case study: <https://8weeksqlchallenge.com/>
 - Data lemur: <https://datalemur.com/>
 - Leetcode: <https://leetcode.com/problemset/database/>

NEXT STEP

- Make full SQL projects and add them to your resume/ LinkedIn/ GitHub profile
- Connect with people on LinkedIn, who are working in data science industry

Hum honge kaamyaab.. 40% completed! 🚀

Week – 6, 7

4. BI Tools (Power BI or Tableau)

Note: *If you are a beginner, my personal suggestion will be to learn Power BI instead of Tableau- as its high in demand and feels similar to MS Excel*

Power BI

- Resources to learn Power BI
 - Power Bi Tutorial + Project Beginners: <https://www.youtube.com/watch?v=6cV3OwFrOkk>
 - Power BI Roadmap: <https://www.youtube.com/watch?v=ZBCAR5Rs7wk>
 - Avi Singh YouTube channel: <https://www.youtube.com/watch?v=AGrl-H87pRU>
 - Power BI Full Course by edureka: <https://www.youtube.com/watch?v=3u7MQz1EyPY>
 - Power BI Udemy course: <https://www.udemy.com/course/powerbi-complete-introduction/>

- Power BI Projects

- **Note:** Below are just examples, do a search on Google and YouTube for more projects-
thoda khud bhi mehnat karo 🤔
- Power BI dashboard project End-to-end: <https://www.youtube.com/watch?v=j4xIVLgsmNQ>
- Power BI dashboard by End-to-End: <https://www.youtube.com/watch?v=et8tAUTwcvY>
- Power BI report by Data With Decision: <https://www.youtube.com/watch?v=0BKlUySopU4>
- End to End Project by KSR datavizon: <https://www.youtube.com/watch?v=aXNhtcQ4nEU>

Tableau

- Resources to learn Tableau

- Tableau Basics For Beginners In Hindi by Great Learning:
<https://www.youtube.com/watch?v=6RZEaEH9ZsQ>
- Tableau by Edureka: <https://www.youtube.com/watch?v=aHaOlvr00So>
- Tableau Tutorial by Simplilearn: <https://www.youtube.com/watch?v=fO7g0pnWaRA>
- Tableau Udemy course: <https://www.udemy.com/course/tableau10/>

- Tableau Projects

- **Note:** Below are just examples, do a search on Google/YouTube for more projects- thoda khud bhi mehnat karo 🤔
- Practice With Examples by Simplilearn: <https://www.youtube.com/watch?v=5uzB4z4iN0g>
- Sales insights by codebasics:
<https://www.youtube.com/watch?v=CCNd2fUfFkk&list=PLeo1K3hjS3usDI9XeUgiNZs6VnE0meBrL>
- Customer Analysis n Dashboard by Stanley:
<https://www.youtube.com/watch?v=qReGTOOrKtk>

Hum honge kaamyaab ek din.. 60% completed! 🚀

5. Programming- Python

Note: If you are a beginner, my personal suggestion will be to **learn Python instead of R**- as its high in demand and beginner friendly. Also, it will help to solve Machine Learning problems

As a beginner learn programming language to an intermediate level, don't waste time to master it 🕒

- Topics in Python:
 - Variables, Data types, Lists, Tuples, Dictionaries, Sets, Conditional expressions, Modules, Functions, Operators, if statements, Loops, classes and objects
 - Python libraries: Pandas and Matplotlib
 - Pandas: read/write csv, excel and JSON files, work with dataframe, data manipulation and analysis- Group by, Concatenate, Merge
 - Matplotlib: creating static, animated, and interactive visualizations in Python
- Resources to learn Python
 - Python Roadmap: <https://www.youtube.com/watch?v=AyMBPxxQRtA>
 - Python Tutorial For Beginners by CodeWithHarry: <https://www.youtube.com/watch?v=gfDE2a7MKjA>
 - Python Course by Udemy: <https://www.udemy.com/course/complete-python-bootcamp/>
 - W3schools website: <https://www.w3schools.com/python/>
 - Python for data science by freecodecamp: <https://www.youtube.com/watch?v=LHBE6Q9Xlzl>
 - Python Tutorial by Mosh: <https://www.youtube.com/watch?v=uQrJOTkZlc>
- Resources to learn Pandas and Matplotlib
 - Python for data science course by Udemy: <https://www.udemy.com/course/python-for-data-science-and-machine-learning-bootcamp/>
 - Pandas Tutorial by CodeWithHarry: <https://www.youtube.com/watch?v=RhEjmHeDNoA>
 - Pandas Tutorials by Krish Naik: <https://www.youtube.com/watch?v=BN0nnnadFI0>
 - Matplotlib Tutorial: <https://www.youtube.com/watch?v=VF5RLjSc8GA>
 - Project- Cricket Data Analytics by Codebasics: <https://www.youtube.com/watch?v=4QkYy1wANXA>
 - Project- Covid Analysis by Simplilearn: <https://www.youtube.com/watch?v=G9NmACvXh8w>

- Solving real world data science tasks with Python Pandas by Keith Galli:
<https://www.youtube.com/watch?v=eMOA1pPVUc4>
- Kaggle free dataset: <https://www.kaggle.com/datasets>

NEXT STEPS

- Make a full project end-to-end using Python, above are just for example you can search more projects on GitHub, Kaggle, YT, Google

Hum honge kaamyaab ek din.. 80% completed! 🚀

Week – 11 & 12

6. Soft Skills

- I. Communication skill
 - II. Analytical skill
 - III. Problem solving skill
 - IV. Story telling
 - V. Business understanding
- Resources to work on soft skills
 - Read latest blogs on data science news, helps you keep updated and improve communication, business understand and storytelling skill
 - Example: towards data science website, news sites, LinkedIn learning, etc
 - Watch podcasts, interviews, industry level projects on YouTube
 - Share your learnings on LinkedIn or other social media platforms

7. Resume & Interview Prep

○ Resume preparation

- Note: There is nothing called PERFECT resume, so keep learning and updating! 📄
- Prepare one page resume and use professional template
- Based on above learnings and projects update your resume
- Also, if you have done any courses/certificates do add them as well
- Tailor your resume based on the role/company you're applying
- Free resume template sites:
 - ✓ novo resume: <https://novoresume.com/resume-templates>
 - ✓ resume io: <https://resume.io/resume-templates>
 - ✓ canva resume: <https://www.canva.com/resumes/templates/>

○ Interview preparation

- Note: Once you have completed all the above steps, just start applying for related jobs.
Giving interview is also a part of your learning 😊
- Be thorough with your resume, even with minute details
- Again, watch podcasts and interview experience shared on YouTube
- Read interview questions available on sites like:
 - ✓ Glassdoor
 - ✓ LinkedIn

All the best- 100% completed! 🚀



DATA analyst jobs: [https://www.glassdoor.co.in/Job/india-data-analyst-fresher-jobs-SRCH_IL.0,5_IN115_KO6, 26.htm](https://www.glassdoor.co.in/Job/india-data-analyst-fresher-jobs-SRCH_IL.0,5_IN115_KO6,26.htm)

Microsoft All Courses:

<https://learn.microsoft.com/en-us/credentials/certifications/data-analyst-associate/?practice-assessment-type=certification>

<https://www.coursera.org/projects/use-power-bi-for-financial-data-analysis-project>

<https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-quickstart-learn-dax-basics>

<https://www.coursera.org/learn/data-driven-decisions-with-power-bi>

<https://learn.microsoft.com/en-us/training/paths/prepare-data-power-bi/>

<https://learn.microsoft.com/en-us/power-bi/create-reports/service-dashboard-create>

Google Data Analytics:

https://www.coursera.org/professional-certificates/google-data-analytics?irgwc=1&utm_medium=partners&utm_source=impact&utm_campaign=4961875&utm_content=b2c&irclickid=UEnSClx9ExyKUEW2I%3A0fE3SjUkHV6MSdqwYIU00

IBM Data Science

https://www.coursera.org/professional-certificates/ibm-data-science?irgwc=1&utm_medium=partners&utm_source=impact&utm_campaign=4961875&utm_content=b2c&irclickid=UEnSClx9ExyKUEW2I%3A0fE3SjUkHV6JTBqwYIU00

Machine Learning with Python:

<https://www.edx.org/learn/machine-learning/massachusetts-institute-of-technology-machine-learning-with-python-from-linear-models-to-deep-learning>

Data Analysis: Statistical Modeling and Computation in Applications

https://www.edx.org/learn/data-analysis/massachusetts-institute-of-technology-data-analysis-statistical-modeling-and-computation-in-applications?index=product&queryID=90509914365286d408ab3e4234e7107b&position=1&results_level=first-level-results&term=MIT+Data+science&objectID=course-124fb823-86b0-4ec1-8892-51496ce3c605&campaign=Data+Analysis%3A+Statistical+Modeling+and+Computation+in+Applications&source=edX&product_category=course&placement_url=https%3A%2F%2Fwww.edx.org%2Fsearch

Introduction to Computer Science and Programming Using Python

<https://www.edx.org/learn/computer-science/massachusetts-institute-of-technology-introduction-to-computer-science-and-programming-using-python>

Fundamentals of Statistics

<https://www.edx.org/learn/statistics/massachusetts-institute-of-technology-fundamentals-of-statistics>

MITx: Supply Chain Analytics

<https://www.edx.org/learn/supply-chain-design/massachusetts-institute-of-technology-supply-chain-analytics>

Probability - The Science of Uncertainty and Data

https://www.edx.org/learn/probability/massachusetts-institute-of-technology-probability-the-science-of-uncertainty-and-data?index=product&queryID=c2588e4454283688b535c0c2e7726431&position=2&linked_from=autocomplete&c=autocomplete

Introduction to Computational Thinking and Data Science

<https://www.edx.org/learn/computer-science/massachusetts-institute-of-technology-introduction-to-computational-thinking-and-data-science>

Fundamentals of Manufacturing Processes

<https://www.edx.org/learn/manufacturing/massachusetts-institute-of-technology-fundamentals-of-manufacturing-processes>

Computational Thinking for Modeling and Simulation

<https://www.edx.org/learn/simulation/massachusetts-institute-of-technology-computational-thinking-for-modeling-and-simulation>